Calibration Console

Select, preview, trim, commit, or cancel

A deterministic value editor with no storage or actuator path

Lesson	033 — project	**Time**	180–240 minutes
Prerequisites	Lessons 031 and 032	**Board**	Arduino Mega 2560
Evidence	TP-X, TP-Y, TP-A, TP-B, LCD, RGB, four LEDs	**Status**	active integration; bench open
Energy class	E1: USB, passive controls, resistor-limited indicators		

Boundary. This project edits two bounded calibration values in volatile memory. It writes no EEPROM and controls no motor, servo, relay, heater, lock, or other actuator. Calibration output is presentation only. Remove USB power to stop physical work; software shutdown is lifecycle behavior, not an emergency stop.

1 Mission and learning outcomes

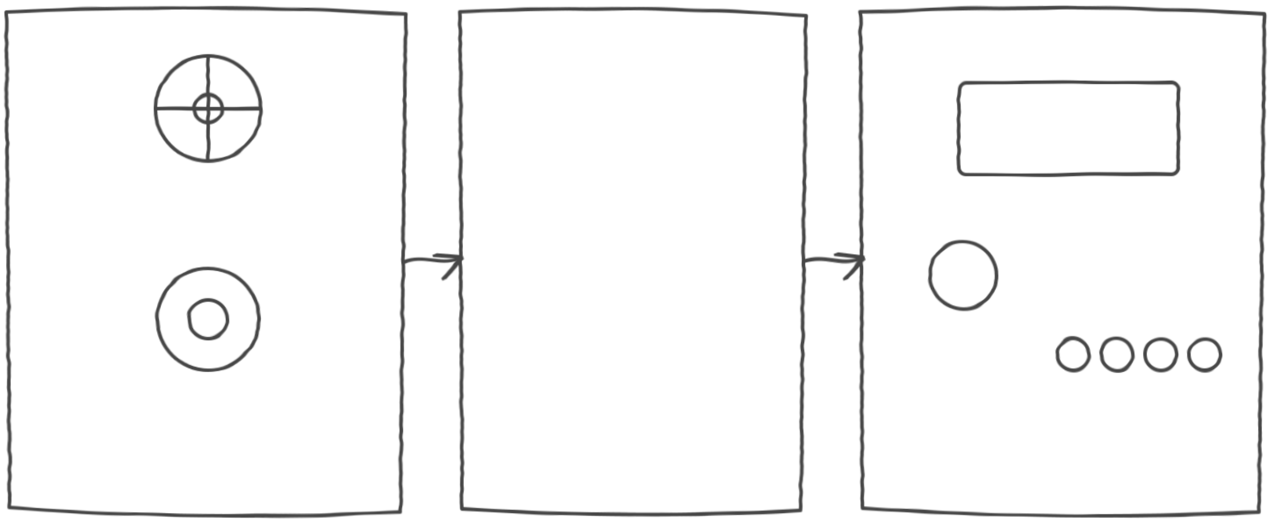
raw controls → interpreted events → bounded preview → explicit commit or cancel.

Lesson 031 supplies raw and interpreted joystick evidence. Lesson 032 supplies valid Gray-code edges, invalid-transition evidence, and a separate encoder button. Lesson 033 coordinates their values in one hardware-neutral console.

After the lesson, you can:

- distinguish a committed pair from a transient preview;
- explain why selection dead bands preserve prior intent;
- apply coarse joystick motion before one-unit encoder trim;
- prove atomic commit and exact cancel rollback with deterministic replay;
- separate raw test-point, component, project, and presentation evidence;
- record acquisition, inactive state, and resource release separately.

2 Monochrome orientation drawing



Joystick + Encoder
TP-X: field
TP-Y: coarse preview
TP-A + TP-B: trim
button: cancel

Mega 2560 + console
supplied-time update
commit/cancel policy

Presentation
LCD: field/values
RGB: console state
four LEDs: preview nibble
optional piezo: acknowledgement

Alternative text: a left-to-right orientation diagram. The joystick provides field selection at TP-X and coarse preview at TP-Y. The encoder provides trim phases at TP-A and TP-B plus a cancel button. A supplied-time console presents field and values on an LCD, state on an RGB LED, and the selected preview's high nibble on four separate LEDs.

This drawing expresses evidence flow, not an exact revision pinout. The qualified Lesson 031 and 032 specimen records and canonical sketch supply the authoritative pin table. Never infer a KY-series marking, pin order, voltage, or pull policy from the retail kit name or this orientation drawing.

Role	Mega pins	Canonical connection
joystick X / TP-X	A0	inspected module X output
joystick Y / TP-Y	A1	inspected module Y output
joystick Select	D22	inspected module switch output
encoder A / TP-A	D24	inspected module A/CLK output
encoder B / TP-B	D25	inspected module B/DT output
encoder Cancel	D26	inspected module switch output
preview nibble	D27–D30	each pin to 1 kΩ, LED anode; cathode to GND
16x2 LCD	D31–D36	RS, EN, D4, D5, D6, D7
common-cathode RGB	D5–D7	one 330 Ω resistor per channel

The exact inspected-module record determines supply, ground, and signal pin order. This table does not authorize an unidentified lookalike.

Point	Observe relative to Mega GND	Interpret only after comparison with
TP-X	horizontal wiper, inside board rails	joystick raw X and selected field
TP-Y	vertical wiper, inside board rails	joystick raw Y and coarse preview
TP-A	encoder A/CLK logic level	ordered A/B phase and encoder snapshot
TP-B	encoder B/DT logic level	ordered A/B phase and encoder snapshot

3 State and value contract

State	Narrow meaning	Circuit-native presentation
Selecting	choose Minimum or Maximum	blue RGB
Editing	change only the selected preview field	amber RGB
Committed	atomic-commit acknowledgement	green RGB
Cancelled	exact-rollback acknowledgement	violet RGB
Fault	invalid input, time, configuration, or invariant	red RGB

The engine owns no endpoint and performs no I/O. The caller supplies time and one input value, then reads a stable snapshot. Its public values are a `TimePoint`, a console input, and a console snapshot; the latter two are named `CalibrationConsoleInput` and `CalibrationConsoleSnapshot`. Construction is inert.

In Selecting, X below -500 chooses Minimum and X above 500 chooses Maximum. The center band preserves the current field. Select copies both committed values to preview and enters Editing.

In Editing, Y outside $-500 \ldots 500$ maps to a coarse candidate across the configured range. Centered Y holds the candidate. Encoder delta is then a one-unit trim. At all times:

$$\begin{aligned} \text{lowerLimit} \le \text{minimum} \le \text{maximum} - \text{minimumSeparation}, \\ \text{minimum} + \text{minimumSeparation} \le \text{maximum} \le \text{upperLimit}. \end{aligned}$$

Select commits both values atomically. Cancel wins over simultaneous Select, restores both preview values, and commits nothing. Acknowledgement lasts the configured duration before returning to Selecting.

Fault rule. Invalid input, backward non-wrapping time, or an impossible configuration enters Fault and commits no value. Recovery requires explicit shutdown and reinitialization. Shutdown clears transient events but leaves the last committed pair inspectable.

4 Narrative Mega adapter

The adapter observes the two components, decides through the value-only console, then presents one snapshot. Presentation cannot call back into the engine.

```
void loop ()
{
    const adk::TimePoint now (millis ());

    observeControls (now);
    decideCalibration (now);
    presentPreview ();
}
```

Setup reads as acquire inputs, acquire indicators, initialize console, show ready. The LCD presents the selected field plus committed and preview values. Four separately resistor-limited LEDs present the high nibble of the selected preview, preserving a non-Serial result if the LCD adapter fails. RGB presents state. A short piezo acknowledgement is optional supporting evidence.

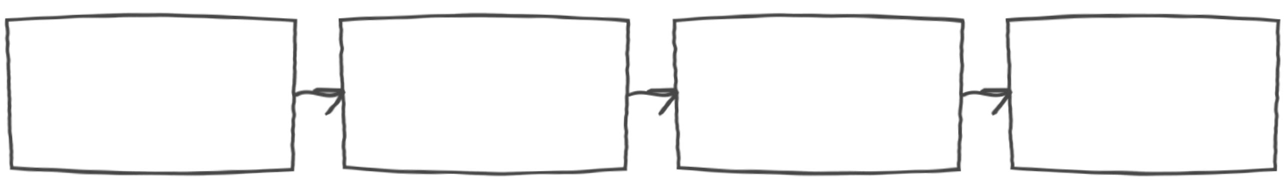
5 Preview, trim, commit, and cancel experiment

1. With USB removed, inspect the exact joystick and encoder revisions. Compare every marking and pin label with their qualified records.

2. Inspect supply and ground, TP-X, TP-Y, TP-A, TP-B, each LED resistor, RGB polarity, and common ground before applying power.
3. Predict Minimum remains selected with centered X. Start the fixture, observe blue, and reconcile TP-X with raw X and the LCD field.
4. Move X past the positive threshold. Predict Maximum, then return X to center and verify that the selection remains stable.
5. Press joystick Select. Predict amber and preview = committed. Move Y beyond its threshold; observe a coarse change on the LCD and four-LED nibble.
6. Turn through one valid encoder edge sequence. Predict one unit of trim. Reconcile TP-A/TP-B order, encoder delta, preview, and presentation.
7. Press Select. Predict one atomic commit and green acknowledgement.
8. Begin another edit and press the encoder button. Predict violet, nothing committed, and exact restoration of both preview values.
9. Present Select and Cancel together. Predict Cancel precedence.
10. Invoke shutdown separately, observe inactive presentation, check resource release with the named method, and then remove USB.

Each payoff depends on the previous state: center hold preserves the selected field, Y supplies the coarse preview, one valid encoder edge supplies one trim unit, Select atomically commits both fields, and Cancel restores both preview values. When Select and Cancel arrive together, Cancel wins and no value is committed.

6 Evidence chain and diagnosis



TP-X / TP-Y / TP-A / TP-B → component snapshots → console snapshot → LCD / RGB / four LEDs

Alternative text: pencil evidence chain from TP-X, TP-Y, TP-A, and TP-B through component snapshots and the console snapshot to the LCD, RGB LED, and four indicator LEDs.

Agreement at one link does not prove the next. An LCD value does not prove the raw voltage; a valid phase sequence does not prove a commit; a dark LED does not prove pin release. Serial output may label an observation, but it is never the only evidence route.

Observation	Stop, inspect, and interpret
field changes while X is centered	compare TP-X, calibration, and thresholds
preview jumps by more than expected	separate Y coarse mapping from encoder trim
encoder count changes on invalid jump	inspect TP-A/TP-B order and invalid counter
only one value restores on Cancel	reject the adapter or engine evidence
red state appears	inspect status; reinitialize only after correcting cause
LED nibble disagrees with LCD	inspect snapshot-to-presentation mapping
out-of-rail test point or hot part	remove USB and troubleshoot unpowered

7 Deterministic replay evidence

Host tests supply every timestamp and input. They cover every state and transition, both fields, center stability, coarse endpoints, every clamp, encoder trim, atomic commit, rollback, simultaneous events, invalid ranges, input failure, acknowledgement boundaries, rollover, restart, and shutdown.

8 The two-minute calibration adventure

The project turns Lessons 031 and 032 into one dependent visible story. Select a field with the joystick, enter Editing, commit one changed preview, begin a second edit, then cancel it with the encoder switch. The LCD preserves the committed and preview values while RGB distinguishes each state and the nibble lamps expose the selected preview. At 120 seconds the sketch shuts down every input and presentation owner regardless of the current state or held input.

9 Claim–evidence–reasoning

Claim	Evidence	What it cannot prove
field selection is correct	TP-X, raw X, field snapshot	endpoint release
trim is one unit	TP-A/B sequence, delta, preview	electrical identity
commit is atomic	before/after pair and state	persistent storage
cancel restores values	committed and preview pair	actuator safety
shutdown is inactive	presentation and pin level	USB power removal

Stop rule. Remove USB power for a hot part, unstable supply, wrong polarity, missing resistor, out-of-rail TP-X/TP-Y/TP-A/TP-B, unexpected output, or disagreement between the qualified pin table and breadboard. Never discover an unknown module rating by experiment.

10 Software evidence and reference

Repository automation owns deterministic replay, lifecycle, sanitizer, Mega, size, PDF, and site gates. These are supporting evidence rather than learner commands, and they do not substitute for physical acceptance.

Reference the ADK `calibration_console.h` interface, focused tests, canonical Lesson 033 sketch, Lessons 031 and 032, the input-expansion design record, and the development, testing, safety, and PDF policies. The searchable HTML lesson is the primary accessible route. This printable PDF uses real text, headings, lists, repeated table headers, and a monochrome drawing, but does not claim PDF/UA conformance.